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In [1]: import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn.preprocessing import LabelEncoder
        from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
        from sklearn.linear_model import LogisticRegression
        from sklearn.tree import DecisionTreeClassifier
        from sklearn.ensemble import RandomForestClassifier

        # Load data
        df = pd.read_csv("student-mat.csv", sep=";")

        # create target variable (pass/fail)
        # Passing = 10 or above, failing = below 10
        df["pass_fail"] = df["G3"].apply(lambda x: 1 if x >= 10 else 0)

        # select features
        # Most relevant features for predicting G3 based on research
        features = ["studytime", "failures", "absences", "G1", "G2"]

        X = df[features]
        y = df["pass_fail"]

        # train/test split
        X_train, X_test, y_train, y_test = train_test_split(
            X, y, test_size=0.2, random_state=42
        )

        # model 1: logistic regression
        log_reg = LogisticRegression(max_iter=1000)
        log_reg.fit(X_train, y_train)

        y_pred_lr = log_reg.predict(X_test)

        print("Logistic Regression")
        print("Accuracy:", accuracy_score(y_test, y_pred_lr))
        print(confusion_matrix(y_test, y_pred_lr))
        print(classification_report(y_test, y_pred_lr))

        # model 2: decision tree classifier
        tree = DecisionTreeClassifier(max_depth=5, random_state=42)
        tree.fit(X_train, y_train)

        y_pred_tree = tree.predict(X_test)

        print("\nDecision Tree")
        print("Accuracy:", accuracy_score(y_test, y_pred_tree))
        print(confusion_matrix(y_test, y_pred_tree))
        print(classification_report(y_test, y_pred_tree))

        # model 3: random forest classifier
        rf = RandomForestClassifier(n_estimators=200, random_state=42)

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rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)

print("\nRandom Forest")
print("Accuracy:", accuracy_score(y_test, y_pred_rf))
print(confusion_matrix(y_test, y_pred_rf))
print(classification_report(y_test, y_pred_rf))

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Logistic Regression

Accuracy: 0.8987341772151899

[[25 2]

[6 46]]

	precision	recall	f1-score	support
0	0.81	0.93	0.86	27
1	0.96	0.88	0.92	52
accuracy			0.90	79
macro avg	0.88	0.91	0.89	79
weighted avg	0.91	0.90	0.90	79

Decision Tree

Accuracy: 0.8987341772151899

[[23 4]

[4 48]]

	precision	recall	f1-score	support
0	0.85	0.85	0.85	27
1	0.92	0.92	0.92	52
accuracy			0.90	79
macro avg	0.89	0.89	0.89	79
weighted avg	0.90	0.90	0.90	79

Random Forest

Accuracy: 0.9113924050632911

[[25 2]

[5 47]]

	precision	recall	f1-score	support
0	0.83	0.93	0.88	27
1	0.96	0.90	0.93	52
accuracy			0.91	79
macro avg	0.90	0.91	0.90	79
weighted avg	0.92	0.91	0.91	79

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